

## Highlights

**The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer**

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

# The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

T. M. Nestor

*Independent researcher, Australia*

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## Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site  $T$ -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, with no modelling or shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's and the lattice kernel at normal incidence is in closed form; (ii) the single-site  $T$ -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form  $(ka)^2$  term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying the first moment of its internal field converges at fourth order, with leading error  $-(kd)^4/720$ , for incident P, SV and SH waves

at normal and oblique incidence. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of  $10^{-6}$  with one voxel per model layer. On a voxelised sphere the error is again second order, dominated by the staircase error. Every result is verified by an independent implementation or an exact solution.

*Keywords:* Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

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## 1. Introduction

*The problem this paper serves.* Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site  $T$ -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the

17 single voxel's response is matched to cancel it, refinement converges to the  
18 wrong answer without revealing it.

19 *Why a homogeneous layer.* This paper therefore starts from the simplest  
20 configuration in which a voxel scheme runs in full (every voxel, every coupling,  
21 the whole multiple-scattering series, at normal and at oblique incidence) and  
22 the exact answer is nevertheless known: a layer filled with identical voxels.  
23 Because cubes tile space, such a layer does not approximate a homogeneous  
24 layer; it *is* one. Any difference between its reflection and transmission and  
25 those of the continuous layer is therefore error of the discretisation alone, with  
26 no modelling error and no shape error behind which to hide. A sphere, the  
27 usual test object for volume-integral and discrete-dipole methods [1], cannot  
28 give this: its error mixes the discretisation with the staircase approximation  
29 of a curved surface by cubes. The layer plays the part of the patch test in  
30 finite elements: reproducing a uniform medium correctly is necessary, though  
31 not sufficient, for a scheme to be trusted in a heterogeneous one. What the  
32 layer establishes about the scheme's two ingredients, the coupling between  
33 voxels and the response of a single voxel, are properties of the ingredients,  
34 which are the same in any medium: that the averaged coupling is exact, that  
35 the self term cancels, and the form of the single-site error. §9 confirms that  
36 they carry over to a stratified medium; a laterally heterogeneous one remains  
37 to be tested.

38 *What the reader gains.* Four things that a heterogeneous test cannot provide.  
39 First, a proof, exact at normal incidence, that the voxel scheme converges  
40 to the continuum, and the specific ingredient, a coupling averaged over the  
41 source voxel, without which it converges to the wrong answer. Second, the

42 leading error in closed form for every discretisation, so that a grid can be  
 43 chosen for a required accuracy before the calculation is run. Third, a single-  
 44 site  $T$ -matrix derived from first principles, with its error stated and checked  
 45 against the exact solution for a sphere. Fourth, a voxel that converges at  
 46 fourth order, and with further Legendre moments at any even order, also in  
 47 a randomly stratified model (§9): for a reflection accuracy of  $10^{-6}$  it allows  
 48 voxels 58 times larger in each direction than the standard one (§11).

49 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*  
 50 *space*, so a uniform region discretised into cubes is still that uniform region,  
 51 whereas spheres or point scatterers can only approximate it and carry a pack-  
 52 ing error of their own. Their *regular lattice* makes the interaction between  
 53 cells translation invariant, so the coupling is a convolution (Toeplitz between  
 54 planes, Bloch-periodic within them), and the whole system is solved by fast  
 55 transforms. And a cube small compared with the wavelength has a *single-site*  
 56 *response in closed form*, the elastic depolarisation of a cube (§6), so the only  
 57 approximation a voxel introduces is the internal field it assumes.

58 *Multiple scattering on a lattice.* Replacing a heterogeneous body by a lattice  
 59 of small scatterers that interact through the background Green's function  
 60 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-  
 61 magnetics its discretised form is the discrete dipole approximation [DDA;  
 62 4, 5], whose accuracy has been studied in detail: each dipole's polarisability  
 63 must account for the local field of the surrounding lattice, which the lattice  
 64 dispersion relation fixes so that an infinite lattice propagates waves as the  
 65 continuum does to a given order in  $kd$  [6]; the error of the method converges  
 66 at second order in the dipole spacing when shape errors are absent [1]; and

67 integrating the Green's tensor over each cell rather than sampling it at the  
 68 centre improves accuracy, most of all for strong contrasts [7]. The elastic  
 69 problem adds a structure the scalar and electromagnetic ones lack. A cubic  
 70 voxel responds to the incident field with a force (through its density contrast)  
 71 and a moment (through its stiffness contrast), so the state at a voxel has nine  
 72 components (displacement and strain), and the single site's self-interaction is  
 73 the elastic depolarisation of a cube, the discrete counterpart of the inclusion  
 74 tensor of Eshelby [8].

75 *This paper.* The approach originates in the two-and-a-half-dimensional for-  
 76 mulation of Nestor [9]: a medium divided into cells that interact through the  
 77 background Green's tensor, each carrying a single-site contrast renormalised  
 78 by its own near field, with the multiple-scattering series summed. This pa-  
 79 per extends it to cubic voxels. §2 sets out the discrete medium and §3 the  
 80 continuous layer it must reproduce. §4 shows why the point coupling fails,  
 81 and §5 why the source-averaged coupling succeeds exactly. §6 derives the  
 82 single-site  $T$ -matrix from moment integrals, tests it against the exact sphere,  
 83 and shows that with it the discrete system is a collocation of the continuum  
 84 equation. §7 gives that collocation's second-order error in closed form, and  
 85 §8 the fourth-order voxel, for incident P and S at normal and oblique inci-  
 86 dence, and its generalisation to Legendre moments of any degree. §9 repeats  
 87 the tests on a randomly stratified layer, §10 on a sphere, and §11 discusses  
 88 what the results imply for voxel schemes in general. Every result is derived  
 89 symbolically in Mathematica and checked against an independent Python  
 90 implementation or an exact solution.

## 91 2. Setting

92 *The discrete medium.* Space is tiled by cubes of side  $d$  (half-side  $h = d/2$ ).  
 93 Cube  $i$  carries a contrast  $\Delta\lambda, \Delta\mu, \Delta\rho$  from an isotropic background with  
 94 Lamé parameters  $\lambda, \mu$ , density  $\rho$ , P and S speeds  $\alpha, \beta$  and P-wave modulus  
 95  $M_P = \lambda + 2\mu$ . Time dependence is  $e^{-i\omega t}$ ;  $k_P = \omega/\alpha$ ,  $k_S = \omega/\beta$ . Coordinates  
 96 are ordered  $(z, x, y)$  with  $z$  down. The field at a voxel is the nine-component  
 97 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

98 displacement followed by engineering strain. A voxel responds with a force  
 99 and a moment (stress dipole); the  $9 \times 9$  single-site  $T$ -matrix maps the exciting  
 100 state to them, and the  $9 \times 9$  lattice Green's tensor  $\mathbf{K}(i, j)$  maps a source  
 101 voxel's force and moment to the state at voxel  $i$ . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

102 with  $\psi^0$  the incident state. Strain rows of  $\mathbf{K}$  are receiver derivatives of the  
 103 point Green's tensor  $\mathbf{G}$ ; moment columns are derivatives with respect to the  
 104 receiver coordinate, symmetrised with a factor  $\frac{1}{2}$  on the shear components.  
 105 In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

106 where  $\Delta\mathcal{C}$  is the  $6 \times 6$  contrast stiffness with  $\Delta\lambda + 2\Delta\mu$  and  $\Delta\lambda$  in the normal  
 107 block and  $2\Delta\mu$  (not  $\Delta\mu$ ) on the shear diagonal.

108 *The laterally uniform layer.* A layer of thickness  $D$  is filled with  $n$  planes  
 109 of voxels,  $d = D/n$ , and illuminated at normal incidence. The field is then

110 laterally uniform, only the zero-lateral-wavenumber (specular) component of  
 111 the lattice coupling acts, and Eq. (2) becomes exactly a chain of  $n$  planes  
 112 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

113 where the sum runs over the lattice vectors  $\mathbf{R} = d(0, p, q)$ ,  $p, q$  integers, that  
 114 join the centres of the voxels in one plane (a square lattice of pitch  $d$ ), and  
 115  $m$  is the separation of the two planes in units of  $d$  (Fig. 1a,b). The question  
 116 is whether, and how fast, this chain converges to the continuous layer as  
 117  $n \rightarrow \infty$  at fixed  $D$ .

118 *Implementations.* Every result is derived symbolically in Mathematica and  
 119 compared with an independent numerical implementation or an exact solu-  
 120 tion. “The package” below is the Python implementation of the voxel scheme  
 121 (`cubic_scattering`): its cube  $T$ -matrix, its lattice kernels and its stratified  
 122 propagator, which serve both as the object under test and, where validated,  
 123 as a second implementation.

### 124 3. The continuous layer and its thin-layer expansion

125 At normal incidence a plane force drives a one-dimensional problem per  
 126 wave type, with modulus  $M$  ( $M_P$  for P,  $\mu$  for S), wavenumber  $k$  and plane  
 127 Green’s function  $g(z) = \frac{i}{2Mk} e^{ik|z|}$ . The layer’s reflection and transmission  
 128 follow from continuity of displacement and traction at its faces; they agree  
 129 with the stratified propagator used as the reference in this work to  $2.5 \times 10^{-8}$ ,  
 130 P and S (`ContinuumLimit_Reference.wl`).

131 For a layer of thickness  $D$  with modulus  $M_1$  and density  $\rho_1$  in a back-  
 132 ground  $M_0, \rho_0$ , write  $Z_j = \sqrt{M_j \rho_j}$  for the impedances,  $\zeta = Z_1/Z_0$  for their  
 133 ratio and  $k_1 = \omega \sqrt{\rho_1/M_1}$  for the layer's wavenumber. The reflection coeffi-  
 134 cient expands as

$$R = \frac{ik_1 D}{2}(\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4}(\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24}(\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48}(\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (5)$$

135 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each  
 136 coefficient symbolically and the truncated series against the exact layer).  
 137 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[ \frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left( \frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (6)$$

138 to first order a thin layer is an interface across which the traction jumps by  
 139 the excess inertia and the displacement by the excess compliance. At oblique  
 140 incidence the same statement holds with the first-order system  $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$   
 141 for the P-SV state  $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$ : the layer acts to  $O(D)$  as the jump  
 142  $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$ , which reproduces the layer's  $O(D)$  reflection coefficients  
 143  $R_{PP}, R_{PS}$  to  $10^{-12}$ ; the series truncated at  $O(D^k)$  converges at exactly order  
 144  $k$ ; and a soft thin layer with shear modulus  $\mu_1 = D/\eta$  at fixed compliance  $\eta$   
 145 tends to the linear-slip interface as  $D \rightarrow 0$ . These expansions are the target  
 146 a single plane of voxels must reproduce order by order. A plane of voxels  
 147 that carry Legendre moments of their internal field to degree  $p$  reproduces  
 148 Eq. (5) term by term through  $O(D^{2p+2})$  and first differs at  $D^{2p+3}$  (§8): the  
 149 mean-only voxel reproduces the equivalent interface and the  $D^2$  term, the  
 150 first-moment voxel all four terms shown.

#### 151 4. The specular sums of the point Green's tensor

152 *Why transform the sum.* The specular sum (4) adds, at a receiver, the field  
 153 of every voxel of a plane  $m$  planes away. In that form it is a poor object to  
 154 study. The Green's tensor decays slowly with distance, so the sum converges  
 155 slowly, and it hides the one question that matters here: which part of it is  
 156 the continuum, and which part is the discretisation. The Poisson summation  
 157 formula [10] answers both at once, and is the standard tool for lattice sums  
 158 of wave fields [11]. For a function  $f$  on the plane and the square lattice of  
 159 pitch  $d$  it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g}\cdot\mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (7)$$

160 with  $p, q$  integers: a sum of samples of  $f$  on the lattice equals a sum of samples  
 161 of its Fourier transform on the reciprocal lattice, whose pitch is  $2\pi/d$  (Fig. 2).  
 162 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (8)$$

163 with  $\hat{\mathbf{G}}(\mathbf{g}; z)$  the two-dimensional (lateral) transform of the Green's tensor  
 164 between planes a distance  $z$  apart. Each term has a physical meaning. Trans-  
 165 formed laterally, the field of a plane of sources is a superposition of plane  
 166 waves, one for each lateral wavenumber, and the lattice selects the wavenum-  
 167 bers  $\mathbf{g}$ . The order  $\mathbf{g} = 0$  is the laterally uniform plane wave that a continuous  
 168 layer radiates: it is the continuum's term. Every other order has  $|\mathbf{g}| \geq 2\pi/d$ ,  
 169 far beyond the wavenumber of the field when the voxels are small compared  
 170 with the wavelength, so it is evanescent and decays away from the plane as

171  $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$ , independently of  $d$ . The transform therefore splits  
 172 the lattice sum exactly into the continuum and a discretisation correction,  
 173 and the correction converges after a few orders.

174 *The local term that survives.* In the strain-from-moment block the evanescent  
 175 orders carry two powers of  $|\mathbf{g}| \propto 1/d$ , so their contribution per voxel volume  
 176 does *not* vanish as  $d \rightarrow 0$ : at  $m = 1$  it is eleven times the continuum  
 177 term (`ContinuumLimit_SpecularSums.wl`, matched to the package’s Ewald  
 178 lattice sum to  $5 \times 10^{-13}$ ). This surviving term is static. Summed over all  
 179 planes in slab order, the static lattice sum of the point kernel is exactly  
 180 the depolarisation of a uniform plate ( $-1/M_P$  on  $\varepsilon_{zz} \leftarrow M_{zz}$  and  $-1/(2\mu)$   
 181 on the normal shears, per unit volume) plus a cubic-symmetric remainder  
 182 (`ContinuumLimit_StaticLocal.wl`, to  $2 \times 10^{-10}$ ). The point kernel therefore  
 183 reproduces the continuum only if that remainder is cancelled by the voxel’s  
 184 own self term.

## 185 5. The tiling identity

186 Replace the point coupling by the *source-cell average*  $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$   
 187  $\mathbf{y}) d\mathbf{y}$ , the field at a voxel centre of a uniform polarisation of the source voxel.  
 188 Each Poisson order in Eq. (8) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (9)$$

189 which vanishes at every  $\mathbf{g} \neq 0$  of the reciprocal lattice since  $\text{sinc}(p\pi) = 0$  for  
 190 integer  $p \neq 0$ . This is the Fourier form of the fact that cubes tile the plane,  
 191  $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$  for the cube’s indicator  $\chi$ : a partition of unity (Fig. 3).  
 192 Two exact statements follow.

193 *Between planes.* For  $m \neq 0$  every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (10)$$

194 the continuum's plane-wave term times the vertical form factor. The pack-  
195 age's cell-averaged kernel agrees in all 81 entries to  $2.8 \times 10^{-16}$  (`ContinuumLimit_`  
196 `AveragedSums.wl`).

197 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,  
198 is the field at the mid-plane of a uniform plate of thickness  $d$  (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (11)$$

199 where  $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$  is elementary: in the one-dimensional  
200 transform each entry is  $c_0 + A/(Mk_z^2 - \rho\omega^2)$ , so it averages to  $c_0/d + A(e^{ikh} -$   
201  $1)/(dMk^2)$ . Its only nonzero entries are  $u \leftarrow f$  on the diagonal and the  
202 three normal strain-from-moment entries, whose local weights  $-1/M_P$  and  
203  $-1/(2\mu)$  are the plate depolarisation of §4. Statically the identity holds for  
204 all 21 independent components of the strain-from-moment tensor to  $4 \times 10^{-9}$ ,  
205 with every cell integral evaluated analytically (below).

206 *Consequence: an exact kernel with no lattice sum.* At normal incidence the  
207 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (12)$$

208 with no Ewald splitting, near shell or asymptotic tail. The route it replaces  
209 (Ewald summation with a near shell averaged by quadrature and an  $O(d^2)$   
210 Taylor tail beyond it) carries a static strain-from-moment error of  $9.4 \times 10^{-4}$

211 at its default settings, falling to  $2.6 \times 10^{-5}$  only when both the shell radius  
 212 and the quadrature order are raised; Eq. (12) is now the package's kernel at  
 213 normal incidence.

214 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as  $G_{bc} =$   
 215  $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$  with  $c_B = (1/\mu - 1/M_P)/8\pi$  and  $c_A = 1/(4\pi\mu) - c_B$ .  
 216 Every cell integral of its derivatives is a cell integral of derivatives of  $1/r$  and  
 217  $r$ . One application of the divergence theorem moves it onto faces  $x_p = c_p \pm h$ ,  
 218 which never contain the origin, and there the two-dimensional antiderivative  
 219 in the face coordinates is continuous, so the face integral is a sum over the  
 220 face's corners. The distributional term at  $r = 0$  is carried by the divergence  
 221 theorem itself. (A triple antiderivative evaluated at the cube's corners is  
 222 *not* usable for repeated derivatives: its arctangent terms jump across the  
 223 coordinate planes through the origin, and differentiating twice there discards  
 224 exactly that term.) The self term so obtained reproduces the classical closed  
 225 form of the cube's depolarisation to machine precision.

## 226 6. The single site: the cube $T$ -matrix from moment integrals

227 Every error of a voxel scheme is made at the single site (§11), so the single  
 228 site is derived here rather than imported. The classical route imports it. For  
 229 an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a  
 230 uniform load uniform, with a depolarisation tensor in closed form [8, 12, 13];  
 231 its dynamic generalisation is known for spheres and ellipsoids [14, 15]. For  
 232 a cube or a general polyhedron the static field of a uniform eigenstrain is  
 233 also in closed form [16, 17, 18, 19], but the strain it produces is not uniform,  
 234 so there is no exact uniform-strain inclusion to import. In scattering theory

235 the same object appears as the self-consistent or effective-field closure of  
 236 the volume integral equation [20, 21, 22]: a first-order (Born) scatterer [23]  
 237 renormalised by the geometric series of its own near field, the step that turns  
 238 a contrast into an effective contrast. Nestor [9] introduced the single site  
 239 in this form, summing the series of the cell's self integral, with the singular  
 240 near field evaluated over an infinitesimal cell. The moments below evaluate  
 241 the same self integral over the finite cube, as a distribution. The elastic  $T$ -  
 242 matrix of Waterman [24] represents a scatterer in spherical multipoles, the  
 243 natural basis for a sphere; a voxel in a lattice needs one compatible with the  
 244 lattice, and the moments over the cube provide it. The construction below  
 245 makes the approximation of a voxel explicit rather than implicit: it derives  
 246 the cube's  $T$ -matrix from the moments of the elastodynamic Green's tensor  
 247 over the cube, and identifies the uniform-strain closure as one truncation of  
 248 an exact hierarchy.

### 249 *6.1. The closed hierarchy*

250 Let the voxel be  $V = [-h, h]^3$  with contrast  $\delta c_{ijkl} = \Delta\lambda\delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$   
 251  $\delta_{il}\delta_{jk})$  and  $\Delta\rho$ . Expand the internal displacement about the centre to second  
 252 degree and project the Lippmann–Schwinger equation onto 1,  $x_p$  and  $x_px_q$ .  
 253 The unknowns are  $u_i(0)$ ,  $\partial_p u_i(0)$  and  $\partial_p \partial_q u_i(0)$ , and the result is the closed

254 system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (13)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (14)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (15)$$

255 all fields evaluated at the centre. Its coefficients are six moments of the  
256 background Green's tensor  $\mathbf{G}$  over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (16)$$

257 with derivatives taken with respect to the source point,  $\partial' = -\partial$ . That sign  
258 reaches  $\mathbf{N}$  and  $\mathbf{Q}$  only, and no symmetry, parity or scaling check can detect  
259 it, so it is fixed by convention. The moments are graded by  $(D, W)$ , the  
260 number of derivatives and the degree of the weight:  $\mathbf{G}(0, 0)$ ,  $\mathbf{K}(0, 2)$ ,  $\mathbf{N}(1, 1)$ ,  
261  $\mathbf{M}(2, 0)$ ,  $\mathbf{P}(2, 2)$ ,  $\mathbf{Q}(3, 1)$ .

## 262 6.2. The moments as distributions

263 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[ \delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (17)$$

264 [25, 26]. Expanding  $g_c$  in powers of  $r$  reduces every moment in Eq. (16) to a  
265 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (18)$$

266 For  $m = -1$  and  $D \geq 2$  the classical integrand is not absolutely integrable.  
 267  $\partial^2(1/r)$  decays as  $r^{-3}$ , its angular average vanishes, and the value of the inte-  
 268 gral depends on the shape of the region excluded around the origin. This is  
 269 the source of the  $\delta$ -function term of the inclusion problem, which a numerical  
 270 quadrature misses. Without it the depolarisation is wrong; for a sphere it  
 271 vanishes altogether (§6.3). It is not added afterwards here. The derivatives  
 272 are distributional from the start, and each is moved onto the voxel's faces by  
 273 pairing with the test function  $w\mathbf{1}_V$ , using  $\partial_q\mathbf{1}_V = -n_q\delta_{\partial V}$ :

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (19)$$

274 The surface term is evaluated on the faces, a distance  $h$  from the origin, where  
 275 every derivative of  $r^m$  is smooth. The recursion lowers  $D$  and  $W$  together  
 276 and ends on convergent integrals. The face integrals are done in closed form  
 277 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p; ] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (20)$$

278 which the face route satisfies exactly and a spherical excision returns as  
 279 0. The closed forms contain logarithms of the units  $(2 \pm \sqrt{3})^n$  of the Pell  
 280 equation  $x^2 - 3y^2 = 1$ . Left unreduced, these cancel catastrophically: for  
 281 example,  $\log(72010600134783751 - 41575339372323900\sqrt{3})$  is a cancellation  
 282 of 35 digits. They are therefore reduced to  $n \log(2 - \sqrt{3})$ , which is exact.  
 283 Because every power of  $r$  is paired this way, the dynamic moments keep the  
 284 even-power terms of the expansion  $(k^2 r, r^3, \dots)$  as well as the odd ones.  
 285 A route that integrates only the polynomial (odd-power) terms of  $\mathbf{G}$  misses  
 286 those terms, at a real  $O((kh)^2)$  cost (§6.4).

### 287 6.3. Parity, and the nine-component block

288 The cube is centrosymmetric. A moment that couples a field quantity of  
 289 degree  $a$  to one of degree  $b$  therefore vanishes unless  $a + b$  is even.  $u$ ,  $\partial u$  and  
 290  $\partial\partial u$  have degrees 0, 1 and 2, so Eq. (14) contains the first gradient alone:  $\partial u$  is  
 291 determined by its own block and couples to neither  $u$  nor  $\partial\partial u$ . The leading  
 292 scattering by a stiffness contrast is the stress dipole  $\int_V \delta c : \nabla u \, dV$ , which  
 293 needs only  $\partial u$ . The leading far field of the stiffness contrast is therefore fixed  
 294 by Eq. (14), and *no treatment of the second-gradient block (15) can change*  
 295 *it*. What Eq. (15) contributes reaches the far field only through the  $O((kh)^2)$   
 296 coupling in Eq. (13). That is the order, and the origin (the variation of the  
 297 field across the voxel), of the collocation error of §7, which the first-moment  
 298 voxel of §8 removes.

299 With  $\Delta\rho = 0$ , Eq. (14) is the  $9 \times 9$  operator  $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$ , built  
 300 from  $\mathbf{M}$  alone. Under  $O_h$  the nine gradient components decompose as  $A_{1g} \oplus$   
 301  $E_g \oplus T_{1g} \oplus T_{2g}$ , and the operator is diagonal in that basis. Its eigenvalues are  
 302 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (21)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (22)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (23)$$

303 The density channel is the displacement block of Eq. (13) with the second  
 304 gradients dropped, an amplification  $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$  with  $\mathbf{G}$  the cube integral  
 305 of  $\mathbf{G}$ .

306 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-  
 307 tional channel  $1 + \Delta K / (\lambda + 2\mu)$ , with  $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$ , and a single deviatoric  
 308 channel  $1 + 2\Delta\mu S_{\text{sph}}$ , with  $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$ . The cube's di-  
 309 latational channel, Eq. (21), is the sphere's exactly: its eigenvalue involves  
 310  $\mathbf{M}$  only through the contraction  $\int_V \partial_i \partial_n G_{in} dV$ , and the static Navier equa-  
 311 tion makes  $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$ , whose integral does not depend on  
 312 the shape. The cube's shear channels differ from each other. Their isotropic  
 313 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (24)$$

314 The  $\sqrt{3}$  terms cancel in the average. What a cube adds to a sphere at this  
 315 order is therefore a pure cubic anisotropy. It splits the deviatoric channel  
 316 about the sphere's value, weighted 3 : 2 by the dimensions of  $T_{2g}$  and  $E_g$ ,  
 317 and the diagonal shear is the more depolarised. At the background of §7  
 318 ( $\nu = 7/32$ ),  $2\mu S_{\text{off}} = 0.4314$ ,  $2\mu S_{\text{diag}} = 0.5929$  and the sphere's value is  
 319 0.4960.

320 *Where Eshelby enters.* Equations (21)–(23) follow from the moment integrals  
 321 and the group theory alone. No ellipsoidal result is used, and no  $\delta$  term is  
 322 added by hand. What corresponds to Eshelby is the *truncation*. Keeping  
 323  $\partial u$  and discarding  $\partial\partial u$  is exactly the uniform-internal-strain ansatz, and by  
 324 the uniformity theorem that ansatz is exact if and only if the inclusion is  
 325 an ellipsoid. For a sphere the construction is therefore exact in the static  
 326 limit, and for a cube it is an approximation. The hierarchy (13)–(15) states  
 327 that approximation as the omitted blocks, rather than as an unquantified  
 328 shape correction. A polynomial expansion about the centre cannot represent

329 the edge and corner concentrations of an isolated cube's internal field at any  
 330 finite degree. A voxel in a uniform layer has no such concentrations: its  
 331 neighbours have the same properties, so its internal field is the continuum's,  
 332 which is smooth. That is why a low-degree internal field suffices in a lattice  
 333 (§§7–8) although it is only an approximation for an isolated cube.

334 *Against the exact sphere.* These statements are tested against the exact elas-  
 335 tic Mie solution for a sphere, with no Eshelby formula in between. The  
 336 concentration of Mie order  $n$  is  $E_n = a_n/a_n^{\text{Born}}$  (orders 0, 1, 2 are the dilata-  
 337 tional, density and shear channels), extrapolated to  $ka \rightarrow 0$ . The formulation  
 338 is applied to a ball of radius  $a$ , with its moment  $\mathbf{M}$  evaluated by the surface  
 339 route of Eq. (19) on  $|\mathbf{r}| = a$ . Its dilatational and shear channels equal Mie's  
 340 to  $2 \times 10^{-10}$ , and the density channel is 1. The cube's closed forms meet  
 341 the same numbers. Its dilatational channel is Mie's  $E_0$  to  $2 \times 10^{-10}$ , and its  
 342 isotropic shear average, Eq. (24), is the depolarisation read from Mie's  $E_2$   
 343 to  $10^{-8}$ . The same moment taken as a principal value, over the ball with  
 344 a small sphere excluded around the origin, has zero depolarisation in every  
 345 channel. The whole static response of the sphere is the  $\delta$  term.

346 At finite frequency the closure uses the dynamic ball moments, which are  
 347 in closed form. It departs from Mie at second order in  $ka$  and first order in  
 348 the contrast, and the departure is closed-form too: the ball's moments are  
 349 elementary, and Mie's low-frequency series with the contrast kept exact gives

350  $E_n$  to relative order  $(ka)^3$ . To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (25)$$

351 with no  $O(ka)$  term and no  $O((ka)^3)$  term. The imaginary  $(ka)^3$  terms,  
 352 the radiation damping, agree exactly in all three channels. So the closure's  
 353 only error at this order is the real  $(ka)^2$  term, the uniform-field ansatz's  
 354 own. That is the order the hierarchy assigns to the omitted second gradi-  
 355 ents (§6.3), and it has the same origin, the variation of the field across the  
 356 scatterer, as the collocation error of §7. Eq. (25) is derived symbolically  
 357 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from  
 358 the numerical Mie solve and an independent implementation of the closure  
 359 to  $5 \times 10^{-4}$ , which is the size of the next order in the contrast.

#### 360 6.4. The collocation closure

361 The collocation closure takes the internal field of a voxel as uniform, the  
 362 truncation of §6.3 for the density and stiffness channels together. The voxel's  
 363 own field at its centre is then  $\mathbf{S}_{\text{self}} \Delta e$  for internal state  $e$ , where  $\mathbf{S}_{\text{self}}$  is the  
 364 nine-component form of the moments  $\mathbf{G}$  and  $\mathbf{M}$ . By parity it is block-diagonal.  
 365 So  $e = \psi + \mathbf{S}_{\text{self}} \Delta e$  and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (26)$$

366 With the kernel (12) the self term cancels. Substituting  $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$   
 367 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (27)$$

368 an identity checked on the matrices to  $4 \times 10^{-16}$ . Every block of  $\mathbf{K}_{\text{all}}$  is an  
 369 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (10), (11)),  
 370 so Eq. (27) is precisely the collocation of the continuum Lippmann–Schwinger  
 371 equation with a strain piecewise constant over voxels. It does not contain  
 372  $\mathbf{S}_{\text{self}}$  at all.

373 The package’s cube  $T$ -matrix is this closure statically: both blocks agree  
 374 with Eq. (26) to  $2 \times 10^{-5}$  at  $k_S h = 0.01$ , the difference falling as  $(kh)^2$   
 375 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about  $0.25(kh)^2$   
 376 through its far-field form-factor corrections. Its self integrals are exact in the  
 377 static and radiation parts but omit the real even-power terms of the Green’s  
 378 tensor’s expansion ( $k^2 r, r^3, \dots$ ), an error of 0.4–0.6  $(kh)^2$  against the analytic  
 379 self term, whose series agrees with the closed-form elastodynamic tensor to  
 380  $10^{-14}$ . By Eq. (27) that error cannot enter the closure scheme.

## 381 7. Convergence, and the closed-form error

382 *Second order.* For a layer  $D = 2\text{ m}$  with the contrast  $\Delta\lambda = 2\text{ GPa}$ ,  $\Delta\mu =$   
 383  $1\text{ GPa}$ ,  $\Delta\rho = 100\text{ kg m}^{-3}$  in a background  $\alpha = 5000$ ,  $\beta = 3000\text{ m s}^{-1}$ ,  $\rho =$   
 384  $2500\text{ kg m}^{-3}$ , illuminated by a P plane force, the reflected displacement of  
 385 the chain converges to the continuous layer’s at order 2.000 for  $n = 4 \rightarrow$   
 386 16 at  $\omega = 60, 300$  and  $600\text{ rad s}^{-1}$ , with the collocation (27) and with the

387 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on  
 388  $k_S h$  alone.

389 *The leading error in closed form.* In Eq. (27) the kernel is integrated exactly;  
 390 the only approximation is the internal field's variation across a voxel. At first  
 391 order in the contrast the discrete scattered field is therefore a midpoint rule  
 392 applied to the continuum's Born integral, with the kernel's cell integral exact.  
 393 For reflection the integrand carries the two-way phase  $e^{2ikz}$ ; for transmission  
 394 it is flat. With  $k$  the background wavenumber of the wave concerned ( $k_P$  for  
 395 an incident P wave,  $k_S$  for an incident S wave),  $R$  the reflection coefficient  
 396 and  $T^{\text{sc}}$  the scattered part of the transmitted field, for *every*  $n$

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (28)$$

397 confirmed for  $n = 1$  to 16 to  $10^{-4}$  of the error itself (`ContinuumLimit_`  
 398 `ErrorConstant.wl`); for an incident S wave, with  $k = k_S$ , the same forms  
 399 and those of Eq. (31) hold to  $4 \times 10^{-4}$  (`ContinuumLimit_IncidentS.wl`, §8).  
 400 At the full contrast the reflection constant is unchanged (0.124 against 1/8);  
 401 the transmission constant grows to  $-0.092$ .

402 *Why a uniform-field single site stops at second order.* The two constants have  
 403 opposite signs, so no scalar correction of  $\mathbf{T}$ , no single  $O((kh)^2)$  factor, can  
 404 cancel both. The package *T*-matrix illustrates it: its form-factor corrections  
 405 reduce the reflection error by a factor 4.45 and *increase* the transmission  
 406 error by 1.7, at every  $n$ . The missing term in Eq. (28) is the first moment  
 407 of the internal field over each voxel, which radiates through the derivative of  
 408 the kernel. A fourth-order scheme must carry it: a single site with a gradient

channel, and a kernel whose cell integrals include the kernel's derivative.

## 8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel  $j$ , with local coordinate  $s \in [-h, h]$  and Legendre functions  $\phi_b(s) = P_b(s/h)$  ( $\phi_0 = 1$ ,  $\phi_1 = s/h$ ,  $\phi_2 = (3s^2/h^2 - 1)/2$ , ...), the field is  $w = \sum_b c_{b,j} \phi_b$ ,  $w = (u, \varepsilon)$  the displacement and strain of the P channel, and the polarisation is  $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$  with  $\Delta M = \Delta \lambda + 2\Delta \mu$ . Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (29)$$

with the same functions on every voxel gives

$$\left( \int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[ \int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (30)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ . Between voxels the double integral is separable, a product of Legendre moments of  $e^{\pm i k z}$ ; within a voxel it is split at  $z' = z$  and the local term contributes  $-\int_i \phi_a \phi_b dz / M_P$ . Every entry is closed-form. Keeping only  $\phi_0$  recovers a second-order scheme; keeping  $\phi_0$  and  $\phi_1$  converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full  $9 \times 9$  system to all printed digits.

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , against the number of voxel planes  $n$ : collocation, the mean-only voxel and the first-moment voxel of Eq. (30). The order is 4.00 at  $\omega = 60$  and  $600 \text{ rad s}^{-1}$  too.

| $n$   | collocation                               | mean only                                 | mean and first moment                       |
|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| 1     | $1.8 \times 10^{-3} / 1.3 \times 10^{-3}$ | $1.2 \times 10^{-3} / 1.8 \times 10^{-3}$ | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   |
| 4     | $1.1 \times 10^{-4} / 8.3 \times 10^{-5}$ | $7.5 \times 10^{-5} / 1.1 \times 10^{-4}$ | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   |
| 16    | $7.0 \times 10^{-6} / 5.2 \times 10^{-6}$ | $4.7 \times 10^{-6} / 6.9 \times 10^{-6}$ | $4.2 \times 10^{-12} / 4.2 \times 10^{-12}$ |
| order | 2 / 2                                     | 2 / 2                                     | 4 / 4                                       |

427 *Higher moments.* Carrying the Legendre moments to degree  $p$  changes nothing in the construction: the moments and the same-cell double integrals of  
428  $P_b(s/h)$  against exponentials are again closed-form, and for  $p \leq 1$  the general-  
429 degree solver reproduces the chain above exactly. The second-moment voxel  
430 ( $p = 2$ ) converges at sixth order and the third-moment voxel ( $p = 3$ ) at  
431 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment`.  
432 `w1`); the second-moment voxel's order is 6.00 at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ .  
433 The errors fall to  $10^{-22}$ , so they are computed in 40-digit arithmetic.

435 *The leading error in closed form.* Without scattering, the Galerkin solution  
436 is the  $L^2$  projection  $\Pi$  of the incident field onto each voxel's basis, and the  
437 kernel is integrated over each voxel exactly; at first order in the contrast the  
438 error is therefore the kernel's cell integral applied to  $\Pi w^0 - w^0$ . The field  
439 varies as  $e^{iks}$  across a voxel and the kernel toward an outside observer as  $e^{iqs}$ ,  
440  $q = +k$  for reflection and  $q = -k$  for transmission, so for every scheme and

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , for voxels carrying Legendre moments to degree  $p$ . The order is the mean over the halvings  $n = 2 \rightarrow 4 \rightarrow 8$ .

| $n$   | first moment ( $p = 1$ )                    | second moment ( $p = 2$ )                   | third moment ( $p = 3$ )                    |
|-------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 1     | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   | $2.8 \times 10^{-11} / 4.2 \times 10^{-11}$ | $1.5 \times 10^{-15} / 1.5 \times 10^{-15}$ |
| 2     | $1.7 \times 10^{-8} / 1.7 \times 10^{-8}$   | $4.4 \times 10^{-13} / 6.5 \times 10^{-13}$ | $6.0 \times 10^{-18} / 6.0 \times 10^{-18}$ |
| 4     | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   | $6.9 \times 10^{-15} / 1.0 \times 10^{-14}$ | $2.4 \times 10^{-20} / 2.4 \times 10^{-20}$ |
| 8     | $6.7 \times 10^{-11} / 6.7 \times 10^{-11}$ | $1.1 \times 10^{-16} / 1.6 \times 10^{-16}$ | $9.2 \times 10^{-23} / 9.2 \times 10^{-23}$ |
| order | 4 / 4                                       | 6 / 6                                       | 8 / 8                                       |

every  $n$

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (31)$$

with  $\Pi$  the centre value for collocation, the cell mean for the mean-only voxel and the projection onto  $\{P_0, \dots, P_p\}$  for the voxel of degree  $p$  (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (30) reproduces the forms for collocation and degrees 0 and 1 at  $n = 1$  to 8 to  $10^{-4}$  of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to  $4 \times 10^{-7}$  at a contrast reduced by  $10^{-6}$ . The constant for degree  $p$  follows from the Legendre expansion  $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$ . The projection drops the terms  $j > p$ , led by  $j = p+1$ , whose coefficient is  $O((kh)^{p+1})$ ; its moment against  $e^{iqs}$  is  $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$ . With  $j_n(x) \approx x^n / (2n+1)!!$  and

Table 3: The leading error of the discrete layer in closed form, Eq. (31): discrete over exact scattered field at first order in the contrast, for every number of voxel planes  $n$ .

|               | reflection                        | transmission                      |
|---------------|-----------------------------------|-----------------------------------|
| collocation   | $1 + (kd)^2/8 + O(d^4)$           | $1 - (kd)^2/24 + O(d^4)$          |
| mean only     | $1 + (kd)^2/12 + O(d^4)$          | $1 - (kd)^2/12 + O(d^4)$          |
| first moment  | $1 - (kd)^4/720 + O(d^6)$         | $1 - (kd)^4/720 + O(d^6)$         |
| second moment | $1 + (kd)^6/100800 + O(d^8)$      | $1 - (kd)^6/100800 + O(d^8)$      |
| third moment  | $1 - (kd)^8/25401600 + O(d^{10})$ | $1 - (kd)^8/25401600 + O(d^{10})$ |
| degree $p$    | $1 + (-1)^p c_p (kd)^{2p+2}$      | $1 - c_p (kd)^{2p+2}$             |

452  $h = d/2,$

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (32)$$

453 and the leading error carries  $(q/k)^{p+1}$ : the same sign in reflection and trans-  
454 mission for odd  $p$ , opposite signs for even  $p$ . The first-moment voxel's leading  
455 error therefore does not depend on the direction in which the field is observed.  
456 Eq. (32) is checked symbolically for  $p = 0$  to 5 (every lower power of  $kd$  van-  
457 ishes) in Mathematica and, independently, in sympy.

458 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is  
459 one limit; thinning a single plane is the other. For one plane of voxels ( $n = 1$ ,  
460  $d = D$ ) the series in  $D$  of the reflection coefficient, computed exactly at the  
461 full contrast, coincides with that of the exact layer, Eq. (5), term by term  
462 through  $D^{2p+2}$  and first differs at  $D^{2p+3}$ , for  $p = 0$  to 3 and for a layer both  
463 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

464 w1). Each further Legendre moment adds two terms of the thin-layer series.

465 *In three dimensions.* The three-dimensional first-moment voxel carries the  
 466 nine-component state with Legendre weights  $\{1, x/h, y/h, z/h\}$ , 36 unknowns  
 467 in all, tested with the same functions and coupled by cell-to-cell double in-  
 468 tegrals of the Green's tensor. For a laterally uniform layer at normal in-  
 469 cidence it reduces *exactly* to Eq. (30) (ContinuumLimit\_Reduction3D.w1).  
 470 The lateral first moments are never excited: the lattice and the incident  
 471 field are symmetric under reflection about every voxel's centre planes, and  
 472  $x/h, y/h$  are odd there. The mean and the  $z$ -moment carry weights that  
 473 are constant laterally, so their lateral form factor  $\text{sinc}(g_x h) \text{sinc}(g_y h)$  van-  
 474 ishes on the reciprocal lattice and the plane sum of the double integrals  
 475 is the plate-to-plate integral in  $z$  alone. Two facts mark where genuinely  
 476 three-dimensional structure begins. The lateral first moment's form fac-  
 477 tor,  $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$ , does not vanish on the recipro-  
 478 cal lattice ( $F_1 = i(-1)^p/p\pi$ ), so the gradient channels couple through the  
 479 evanescent orders. And at oblique incidence the mean's own zeros shift,  
 480  $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$ , so the tiling identity holds  
 481 only to  $O(k_{\parallel} h)$ .

482 *Oblique incidence.* The scheme is built spectrally. By Poisson summation  
 483 the Bloch coupling between planes  $i$  and  $j$  at lateral wavenumber  $\mathbf{k}_{\parallel}$  is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g},$$

(33)

484 with  $L_a$  the lateral Legendre factors. The whole-space tensor decays as  
 485  $1/k_z^2$  and the nine-component state adds at most two powers of  $k_z$ , so each

entry of  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$  is a  $\delta(z)$  of constant weight plus, per mode,  $[\mathrm{i}\alpha/2\gamma + \mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$ ,  $\gamma^2 = k_W^2 - \kappa^2$ : every  $z$  integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at  $|p|, |q| \leq 8$ . Against the exact layer, for a P wave incident at  $20^\circ$  and  $40^\circ$  (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P ( $R_{PP}$ ) and converted S ( $R_{PS}$ ) waves for a P wave incident at  $20^\circ$ , with the mean-only and first-moment voxels of Eq. (33). At  $40^\circ$  the orders are again 2 and 4.

| $n$   | mean only            |                      | mean and first moments |                       |
|-------|----------------------|----------------------|------------------------|-----------------------|
|       | $R_{PP}$             | $R_{PS}$             | $R_{PP}$               | $R_{PS}$              |
| 1     | $9.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$ | $7.2 \times 10^{-8}$   | $4.1 \times 10^{-7}$  |
| 2     | $2.3 \times 10^{-4}$ | $4.2 \times 10^{-4}$ | $4.5 \times 10^{-9}$   | $2.5 \times 10^{-8}$  |
| 8     | $1.5 \times 10^{-5}$ | $2.7 \times 10^{-5}$ | $1.7 \times 10^{-11}$  | $9.9 \times 10^{-11}$ |
| order | 2                    | 2                    | 4                      | 4                     |

second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at  $O(k_{\parallel}h)$  and the evanescent coupling of the gradient channels do not cost an order. At  $\mathbf{k}_{\parallel} = 0$  the construction reproduces Eq. (30) to all printed digits.

*Incident S.* Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (30) with  $M_P \rightarrow \mu$ ,  $k_P \rightarrow k_S$  and  $\Delta M \rightarrow \Delta\mu$ ; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full  $9 \times 9$  construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique inci-

501 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders  
are those of Table 5. The orders, and with  $k = k_S$  the closed-form constants

Table 5: Observed order of convergence ( $n = 2 \rightarrow 8$ ,  $\omega = 300 \text{ rad s}^{-1}$ ) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle  $\arcsin(\beta/\alpha) = 36.9^\circ$ .

|                                            | mean only   | mean and first moments |
|--------------------------------------------|-------------|------------------------|
| SV at $20^\circ$ : $R_{SS} / R_{SP}$       | 2.00 / 2.00 | 4.00 / 3.99            |
| SV at $30^\circ$ : $R_{SS} / R_{SP}$       | 1.98 / 2.00 | 3.97 / 4.00            |
| SH at $20^\circ$ and $40^\circ$ : $R_{SH}$ | 2.00        | 4.00                   |

502

503 of Eqs. (28) and (31), are those of the incident P wave.

504 The spectral construction is implemented twice. The second implementa-  
505 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-  
506 nition of the scheme: it takes the depth dependence of the transformed kernel  
507 from an independent residue construction, obtains the local weight as a nu-  
508 merical large- $k_z$  limit of the inverted Christoffel matrix, evaluates every dou-  
509 ble integral by quadrature and separates P from S by sampling the upgoing  
510 field at two depths. The raw reflected displacements of the two agree to  $9 \times$   
511  $10^{-16}$  in all 36 cases compared: incident P, SV and SH, normal incidence and  
512  $20^\circ$ ,  $n = 1, 2, 4$ , with and without the first moments. The one-dimensional  
513 scheme of any degree is also implemented twice. The second implementation  
514 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral  
515 by Gauss quadrature, split at the receiver within a voxel, and the exact layer  
516 from its own interface system, in double precision. Its errors agree with the  
517 40-digit ones within its round-off floor ( $3 \times 10^{-14}$  in reflection,  $1.6 \times 10^{-13}$  in

518 transmission); at  $\omega = 1200 \text{ rad s}^{-1}$  it measures orders 4.02 / 3.99 and 6.02 /  
 519 6.00; for one plane as the layer thins its differences from the exact layer fall  
 520 as  $D^{3.00}$ ,  $D^{5.00}$  and  $D^{7.00}$  for  $p = 0, 1$  and  $2$ ; and it rederives  $c_p$  for  $p \leq 5$ .  
 521 Figure 5(a,b) shows the convergence, with the orders fitted to it.

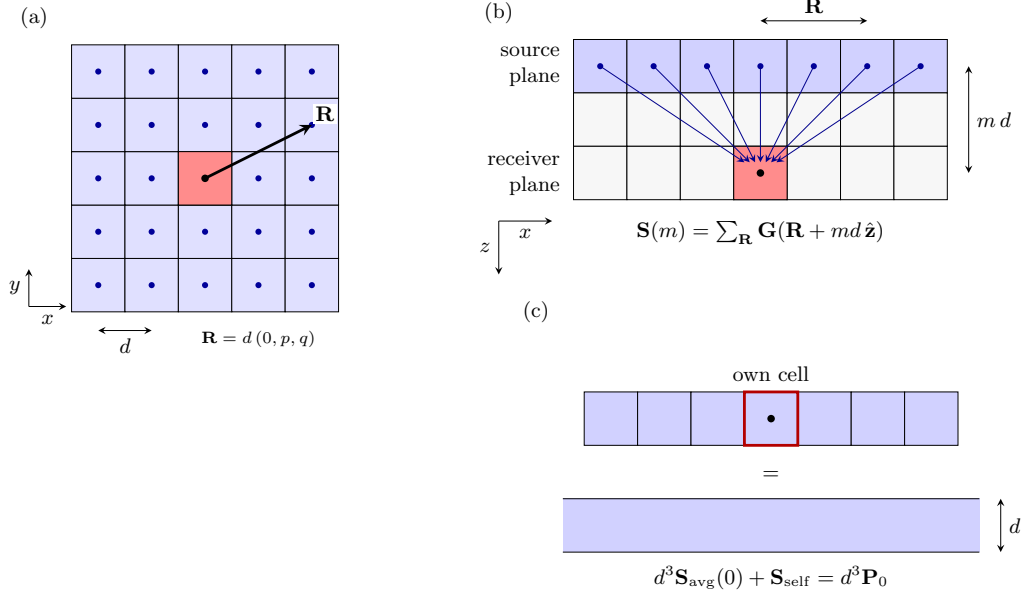


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch  $d$ , and a lattice vector  $\mathbf{R} = d(0, p, q)$  joins the centre of one voxel (red) to another. (b) Cross-section ( $z$  down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane  $m$  planes above it, one for each lattice vector  $\mathbf{R}$ ; the sum over  $\mathbf{R}$  is the specular lattice sum of Eq. (4). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness  $d$ , the tiling identity of Eq. (11).

*Alt text:* Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled  $\mathbf{R}$  runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked  $d$ . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation  $m d$  marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness  $d$ , a uniform plate.

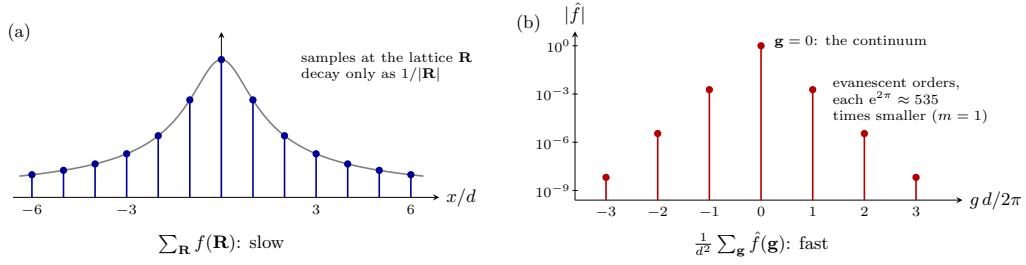


Figure 2: The Poisson summation formula (7) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as  $1/|\mathbf{R}|$ , so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders  $\mathbf{g} = 2\pi p/d$ . The order  $\mathbf{g} = 0$  is the continuum's plane wave; every other order is evanescent and, at  $m = 1$ , smaller than its neighbour by  $e^{2\pi} \approx 535$ . Magnitudes are schematic: the static decay  $e^{-|\mathbf{g}|md}$  is shown.

*Alt text:* Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

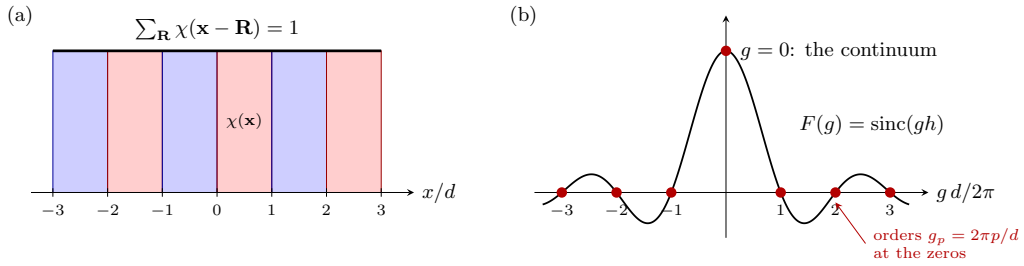


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions  $\chi$  of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor  $F(g) = \text{sinc}(gh)$  (shown along one axis) vanishes at every reciprocal-lattice order  $g_p = 2\pi p/d$  with  $p \neq 0$ , so of all the orders in the lattice sum (8) only  $g = 0$ , the continuum's plane wave, survives the source-cell average.

*Alt text:* Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of  $2\pi/d$ ; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

## 522 9. A heterogeneous layer

523 A heterogeneous medium is not a layer of identical cells. The simplest  
 524 heterogeneous case with an exact answer is a stratified model built from layers  
 525 of constant properties, with real interfaces between them: the form of many  
 526 layered crustal velocity models. The last test takes such a model. A layer  
 527  $D = 4$  m thick is divided into eight model layers of 0.5 m, and in each one  $\Delta\lambda$ ,  
 528  $\Delta\mu$  and  $\Delta\rho$  are drawn uniformly within  $\pm 15\%$  of the background (the values  
 529 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random  
 530 stratification with physical contrasts. The voxels refine the model's layering,  
 531 with  $m$  planes of voxels in each model layer, so no voxel straddles a model  
 532 interface: the natural discretisation when the model's interfaces are known.  
 533 At  $\omega = 1500 \text{ rad s}^{-1}$  the layer is  $k_P D = 1.2$  and  $k_S D = 2.0$  thick, so the field  
 534 reverberates between the model interfaces. The exact answer is the product  
 535 of the model layers' propagators. Its same-type reflection coefficients agree  
 536 with those of the reflection–transmission recursion of Kennett [27], computed  
 537 independently, to  $10^{-14}$ . The schemes are those of §§7–8, with the contrast  
 538 now changing from plane to plane; nothing else changes. The errors are in  
 539 Table 6 and Fig. 5(c).

540 *The interfaces cost nothing.* Every order is the one measured for the uniform  
 541 layer. At a model interface the strain of the true field jumps. A voxel basis  
 542 that is discontinuous from voxel to voxel represents that jump exactly when  
 543 the interface lies on a voxel boundary, so the interface adds no error of its  
 544 own. The only requirement is that voxel boundaries coincide with the model's  
 545 interfaces. A voxel that straddled an interface would carry a contrast that is  
 546 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for  $m$  planes of voxels per model layer (§9). At normal incidence the order is measured from  $m = 4$  to 16, where it is also 2.00 for collocation and 4.00 in transmission; at  $20^\circ$  it is measured from  $m = 2$  to 4.

|                          |               | $m = 1$              | $m = 2$              | $m = 4$              | order |
|--------------------------|---------------|----------------------|----------------------|----------------------|-------|
| P, normal, $R$           | mean only     | $1.9 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $8.5 \times 10^{-7}$ | $5.3 \times 10^{-8}$ | $3.3 \times 10^{-9}$ | 4.00  |
| S, normal, $R$           | mean only     | $5.3 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $6.2 \times 10^{-6}$ | $3.9 \times 10^{-7}$ | $2.4 \times 10^{-8}$ | 4.00  |
| P, $20^\circ$ , $R_{PP}$ | mean only     | $1.4 \times 10^{-3}$ | $3.6 \times 10^{-4}$ | $9.0 \times 10^{-5}$ | 2.00  |
|                          | first moments | $3.3 \times 10^{-7}$ | $2.1 \times 10^{-8}$ | $1.3 \times 10^{-9}$ | 4.00  |
| P, $20^\circ$ , $R_{PS}$ | mean only     | $2.8 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | 2.00  |
|                          | first moments | $1.4 \times 10^{-6}$ | $8.8 \times 10^{-8}$ | $5.5 \times 10^{-9}$ | 4.00  |

547 *What fourth order buys.* For a reflection error of  $10^{-6}$  at normal incidence  
548 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model  
549 layer. The first-moment voxel needs 8, one per model layer: the model's own  
550 layering, which already gives  $8.5 \times 10^{-7}$ . Counting unknowns (9 per voxel  
551 against 36), that is 3123 against 288 per vertical column of voxels, a factor  
552 of eleven. In three dimensions, with the lateral grid refined as the vertical  
553 one is, the number of voxels falls by  $(347/8)^3 \approx 8 \times 10^4$  and the number of  
554 unknowns by about  $2 \times 10^4$ . The same model has been solved independently  
555 in Python, with a contrast for each plane: the raw reflected displacements  
556 agree with the Mathematica scheme to  $1.1 \times 10^{-15}$  at normal incidence and  
557 at  $20^\circ$ , with and without the first moments. Lateral heterogeneity needs the  
558 full three-dimensional solver; it is the first extension discussed below.

## 559 10. A sphere: what the layer predicts in three dimensions

560 The layer and the stratified layer are laterally uniform. A sphere is not,  
561 and it has an exact solution for elastic scattering [28, 29]: it is the usual  
562 benchmark for volume-integral and discrete-dipole methods. This section  
563 solves it with the full three-dimensional voxel scheme, to test what the layer  
564 predicts for a laterally heterogeneous body. The sphere is voxelised by the  
565 cubes of an  $n_{\text{sub}}^3$  grid whose centres lie inside it, and the Foldy–Lax system  
566 is solved with the uniform-field voxel (the second-order scheme of §§6–7;  
567 the first-moment voxel of §8 has not been built for finite three-dimensional  
568 scatterers) and the cell-averaged coupling, by an FFT-accelerated iterative  
569 solver. The single site is the package’s cube  $T$ -matrix, which differs from the  
570 collocation closure only through its far-field corrections (§6.4) and has the  
571 same order. A P wave is incident; the contrast is that of §7. The error is the  
572 largest difference between the scattered far field and the exact one over nine  
573 scattering angles from  $11^\circ$  to  $169^\circ$ , relative to the largest exact value, at a  
574 distance of  $5 \times 10^4$  radii where the far field has converged (`scripts/pilot_`  
575 `sphere_voxel_vs_mie.py`).

576 *The raw error does not converge.* Table 7 shows the error for grids of 4, 6 and  
577 8 voxels across. It does not fall steadily: at  $k_S a = 0.5$  it is 6.1%, then 17%,  
578 then 2.4%. The staircase changes with every grid, and with it the volume and  
579 the shape of the scatterer. Rescaling by the ratio of the sphere’s volume to  
580 the staircase’s removes the largest swings but leaves an error of about 2% at  
581  $k_S a = 0.5$ , and 8 to 11% at  $k_S a = 1.0$ , that does not decrease. A refinement  
582 study of the voxelised sphere measures the staircase, not the scheme.

Table 7: A sphere voxelised on grids of  $n_{\text{sub}}$  voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

| $n_{\text{sub}}$ | voxels | $k_S a = 0.5$         |                      | $k_S a = 1.0$         |                       |
|------------------|--------|-----------------------|----------------------|-----------------------|-----------------------|
|                  |        | raw                   | volume-rescaled      | raw                   | volume-rescaled       |
| 4                | 32     | $6.1 \times 10^{-2}$  | $1.8 \times 10^{-2}$ | $1.19 \times 10^{-1}$ | $7.7 \times 10^{-2}$  |
| 6                | 136    | $1.71 \times 10^{-1}$ | $2.8 \times 10^{-2}$ | $6.8 \times 10^{-2}$  | $1.12 \times 10^{-1}$ |
| 8                | 280    | $2.4 \times 10^{-2}$  | $2.1 \times 10^{-2}$ | $4.6 \times 10^{-2}$  | $8.6 \times 10^{-2}$  |

583 *Refining a fixed shape.* The staircase is itself a union of cubes, so it has its  
584 own exact solution, and the voxel scheme converges to it when each voxel  
585 is split into  $m^3$  sub-voxels of the same shape (Fig. 4). The change of the  
586 far field under that refinement is the discretisation error alone; the differ-  
587 ence between the converged staircase field and the sphere’s is the shape error  
588 alone. Table 8 refines the  $n_{\text{sub}} = 8$  staircase to  $m = 2$  and 3, up to 7560  
589 voxels (`scripts/pilot_sphere_staircase_refinement.py`). The discreti-  
590 sation error converges at order 1.99 at  $k_S a = 0.5$  and 2.06 at  $k_S a = 1.0$   
591 (Fig. 5d): the second order of the uniform-field voxel on the layer, now on  
592 a laterally heterogeneous body. At the original grid it is  $5.4 \times 10^{-4}$  and  
593  $2.7 \times 10^{-3}$ , larger by a factor of about five when the frequency doubles, close  
594 to the factor four of its  $(kh)^2$  scaling. The shape error, from the field ex-  
595 trapolated to  $m \rightarrow \infty$ , is  $2.36 \times 10^{-2}$  and  $4.83 \times 10^{-2}$ : 44 and 18 times the  
596 discretisation error. On a sphere the staircase dominates the error of the  
597 scheme, which is why the layer, where the staircase is exact, is the test that  
598 measures the scheme itself.

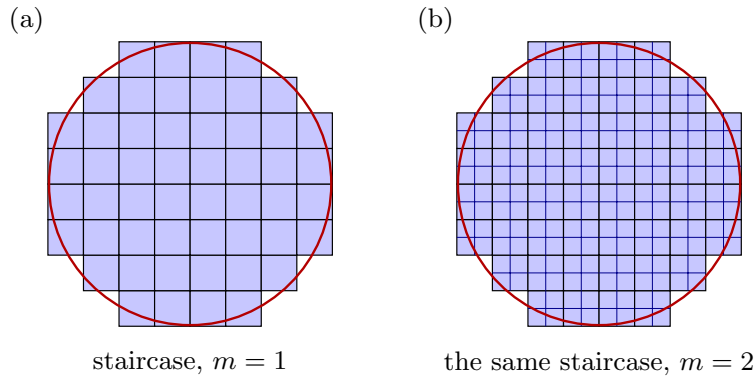


Figure 4: Refining a fixed shape. (a) A cross-section through the  $n_{\text{sub}} = 8$  voxelisation of a sphere (red circle): the voxels whose centres lie inside it. (b) The same staircase with every voxel split into  $2 \times 2 \times 2$  sub-voxels ( $m = 2$ ): the shape, and so its exact solution, is unchanged, and only the discretisation is refined.

*Alt text:* Two panels, each a square grid of shaded cells approximating a circle drawn in red; the shaded cells form a stepped outline that crosses the circle. In the right panel every shaded cell is divided into four smaller cells by thin lines, while the stepped outline is exactly the same as on the left.

Table 8: The  $n_{\text{sub}} = 8$  staircase with every voxel split into  $m^3$  sub-voxels: far-field error against the exact sphere, and the change from the previous level. The change is the discretisation error; its order is measured from the two changes, and the field extrapolated to  $m \rightarrow \infty$  gives the shape error.

| $m$                                    | voxels | $k_{\text{sa}} = 0.5$  |                       | $k_{\text{sa}} = 1.0$  |                       |
|----------------------------------------|--------|------------------------|-----------------------|------------------------|-----------------------|
|                                        |        | vs sphere              | change                | vs sphere              | change                |
| 1                                      | 280    | $2.403 \times 10^{-2}$ |                       | $4.604 \times 10^{-2}$ |                       |
| 2                                      | 2240   | $2.369 \times 10^{-2}$ | $4.04 \times 10^{-4}$ | $4.779 \times 10^{-2}$ | $2.05 \times 10^{-3}$ |
| 3                                      | 7560   | $2.364 \times 10^{-2}$ | $7.51 \times 10^{-5}$ | $4.811 \times 10^{-2}$ | $3.65 \times 10^{-4}$ |
| order of the discretisation error      |        | 1.99                   |                       | 2.06                   |                       |
| shape error ( $m \rightarrow \infty$ ) |        | $2.36 \times 10^{-2}$  |                       | $4.83 \times 10^{-2}$  |                       |

## 599 11. Discussion

600 Figure 5 collects the convergence of every test, with the order fitted to  
 601 each by least squares (`scripts/plot_convergence_orders.py`).

602 *Where the error of a voxel scheme lives.* The results separate the two ap-  
 603 proximations a voxel scheme makes. The *coupling* between voxels makes  
 604 none: averaged over the source cell it is the continuum’s cell integral ex-  
 605 actly, between planes by the form factor’s zeros and within a plane by the  
 606 tiling identity, and the self term it omits is exactly the one the single site  
 607 carries. Every error therefore comes from the *single site*, from the internal  
 608 field it assumes. For a uniform field that error is the midpoint rule’s, known  
 609 in closed form; for a linear field it is the next term of the same expansion.  
 610 The converse is the practical warning. A kernel that is not the cell integral  
 611 (the point Green’s tensor, or a cell average that does not match the single  
 612 site’s closure) leaves a local term that does not vanish as the voxels shrink  
 613 (§4): the evanescent lattice orders carry a static contribution of fixed size per  
 614 voxel volume. A scheme built that way converges to the wrong limit, and  
 615 refinement cannot reveal it, because the error is scale-invariant.

616 *What the single site is.* The moment hierarchy of §6 turns the uniform-field  
 617 single site from a modelling choice into a derived object with a stated error.  
 618 Its static part is exact for a sphere, and for a cube it differs from the sphere’s  
 619 only by the cubic split of Eq. (24). Its dynamic part departs from exact  
 620 scattering by a sphere only through the real  $(ka)^2$  term of Eq. (25); the  
 621 radiation damping is exact. That term and the collocation error of Eq. (28)  
 622 have the same origin, the variation of the field across the scatterer, and the

623 same remedy, a richer internal field. In a lattice the remedy is cheap: the  
 624 field a voxel must represent is the smooth continuum field, not the field of  
 625 an isolated cube, with its concentrations at the edges and corners.

626 *Relation to the discrete dipole approximation.* The same partition appears in  
 627 the electromagnetic literature as the choice of polarisability. The Clausius–  
 628 Mossotti relation, the lattice dispersion relation [6] and the integrated Green  
 629 tensor [7] all adjust the single site so that the lattice reproduces the con-  
 630 tinuum. For space-filling cubes the adjustment is neither needed nor a free  
 631 parameter: the tiling identity fixes the local field exactly, the self term of  
 632 the collocation closure is the one that makes it cancel, and what is left is a  
 633 closed-form error rather than an empirical one. The second-order convergence  
 634 reported for the DDA by Yurkin et al. [1] corresponds to the uniform-field  
 635 voxel’s, and the opposite-signed constants of Eq. (28) explain why a polaris-  
 636 ability correction can improve one observable only at the expense of another,  
 637 as the package’s own form-factor corrections do in reflection and transmis-  
 638 sion. Going beyond second order requires a richer internal field, which is the  
 639 point of higher-order bases for volume integral equations; the first-moment  
 640 voxel is the lowest such basis, and in its spectral form the Galerkin double  
 641 average makes the lattice sum converge without Ewald splitting.

642 *The cost of fourth order.* A first-moment voxel carries four times as many  
 643 unknowns as a uniform one (36 against 9 in three dimensions). The closed  
 644 forms show what that buys. To reach an error  $\varepsilon$  in reflection, the uniform  
 645 voxel needs  $kd = \sqrt{8\varepsilon}$  and the first-moment voxel  $kd = (720\varepsilon)^{1/4}$ ; at  $\varepsilon = 10^{-6}$   
 646 that is  $kd = 2.8 \times 10^{-3}$  against 0.16, voxels 58 times larger in each direction.  
 647 In a three-dimensional model the voxel count falls by  $58^3 \approx 2 \times 10^5$  and the

648 number of unknowns by about  $5 \times 10^4$  (these are the Born-order constants;  
 649 at the contrast studied the first-moment voxel's measured errors lie within  
 650 5% of them, while the uniform voxel's transmission constant grows with the  
 651 contrast, so the comparison is if anything conservative). The stratified model  
 652 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-  
 653 moment voxels for the same accuracy. Higher moments continue the trend  
 654 in one dimension: by Eq. (32) the second-moment voxel reaches  $\varepsilon = 10^{-6}$  at  
 655  $kd = 0.68$  and the third-moment voxel at  $kd = 1.5$ , where a voxel is already  
 656 a quarter of a wavelength. A three-dimensional voxel of degree  $p$  has not  
 657 been built; with the Legendre polynomials of total degree at most  $p$  it would  
 658 carry 10 ( $p = 2$ ) or 20 ( $p = 3$ ) weights per state component, against 4 for  
 659 the first-moment voxel. The gain is largest precisely where it matters: when  
 660 the model varies on scales that are coarse compared with the accuracy the  
 661 wavefield must have.

662 *Limitations and outlook.* The results concern laterally uniform media, a ho-  
 663 mogeneous layer and a randomly stratified one, for which the discrete system  
 664 is exactly a chain of planes; a laterally heterogeneous medium needs the first-  
 665 moment voxel in a full three-dimensional solver, where the gradient channels  
 666 couple through the evanescent lattice orders (§8) and the tiling identity holds  
 667 only to  $O(k_{\parallel}h)$ . The sphere of §10 shows that the second order of the uniform-  
 668 field voxel carries over to a laterally heterogeneous body; the fourth order  
 669 has still to be shown there. The numerical evidence is at moderate, physi-  
 670 cally typical contrasts (up to 15% in density and moduli) and at  $k_S h \leq 0.3$ ,  
 671 the range in which the single-site  $T$ -matrix is validated; the closed forms are  
 672 first order in the contrast, and the departures from them at the full contrast

673 are of second order. What is established is the *order* of convergence and  
 674 its leading constant, not the behaviour at strong contrast, where the single  
 675 site's own closure may limit the accuracy before the lattice coupling does.  
 676 The single site's dynamic error is in closed form for a sphere only; for a cube,  
 677 for which no exact solution exists, it is known through the layer (§§7–8).  
 678 Two extensions follow directly. The first is the three-dimensional solver it-  
 679 self. The second is a stratified *background*, a layered reference medium, as  
 680 distinct from the stratified contrast of §9. There the lattice coupling then  
 681 carries the reverberation of the layering, which is not translation invariant in  
 682 depth, and whether a tiling identity holds for it is the natural next question.

## 683 12. Conclusions

684 A lattice of cubic voxels reproduces a continuous elastic medium in a  
 685 precise sense. Because cubes tile the plane, the source-cell-averaged cou-  
 686 pling between voxels is the continuum's cell integral exactly, and at normal  
 687 incidence the lattice kernel is available in closed form with no lattice sum.  
 688 The single site is derived rather than assumed. The moments of the Green's  
 689 tensor over the cube, taken as distributions, give a closed hierarchy whose  
 690 first-gradient truncation is the uniform-strain closure. That closure is exact  
 691 for a sphere in the static limit, and it departs from exact scattering by a  
 692 sphere only by a real  $(ka)^2$  term, with the radiation damping exact. For  
 693 a cube it differs from the sphere only by a cubic anisotropy of the shear  
 694 response. The collocation closure carries the same self term as the lattice  
 695 kernel, which cancels, so the discrete system is a collocation of the contin-  
 696 uum equation. It converges at second order, with a leading error that is a

697 midpoint-rule error of the internal field, known in closed form for every dis-  
 698 cretisation, and opposite in sign in reflection and transmission. A voxel that  
 699 also carries the first moment of its internal field converges at fourth order,  
 700 at normal incidence (where the three-dimensional scheme reduces exactly to  
 701 one dimension) and at oblique incidence, with a leading error,  $-(kd)^4/720$ ,  
 702 that no longer depends on the direction of observation. With Legendre mo-  
 703 ments to degree  $p$  it converges at order  $2p + 2$  with the closed-form constant  
 704 of Eq. (32), and a single plane of such voxels reproduces the thin-layer se-  
 705 ries term by term through  $O(D^{2p+2})$ . The orders survive the interfaces of  
 706 a randomly stratified model, provided voxel boundaries coincide with its in-  
 707 terfaces, and there the first-moment voxel reaches an accuracy of  $10^{-6}$  with  
 708 one voxel per model layer, where the uniform-field voxel needs about forty.  
 709 On a voxelised sphere the discretisation error is again second order, and the  
 710 staircase approximation of the surface dominates it, by factors of 18 to 44.  
 711 The results, and the evidence for each, are collected in Table 9.

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 714 the Research School of Earth Sciences of the Australian National University,  
 715 Canberra [9].

## 716 Declaration of competing interest

717 The author declares no known competing financial interests or personal  
 718 relationships that could have appeared to influence the work reported in this  
 719 paper.

Table 9: The results of the paper, and the evidence for each.

| Result                                                     | Evidence                                                          |
|------------------------------------------------------------|-------------------------------------------------------------------|
| Continuous layer, exact and to $O(D^k)$                    | reference to $2.5 \times 10^{-8}$ ; series order $k$              |
| Thin-layer series, Eq. (5)                                 | each coefficient symbolically, to $D^4$                           |
| Point-kernel local term survives $d \rightarrow 0$         | $11 \times$ the continuum term at $m = 1$                         |
| Averaged coupling between planes = continuum $\times$ sinc | $2.8 \times 10^{-16}$ , 81 entries                                |
| Tiling identity, Eq. (11)                                  | 21/21 components to $4 \times 10^{-9}$                            |
| Exact kernel, Eq. (12)                                     | replaces a $9.4 \times 10^{-4}$ static error                      |
| Moments as distributions, Eq. (19)                         | sum rule (20) exact; M, Q equal an independent archive, 81/81     |
| Channels (21)–(23)                                         | a production code by another route, $2 \times 10^{-15}$           |
| Cube vs sphere, Eq. (24)                                   | symbolic; and numerically at five $\nu$ , $4 \times 10^{-16}$     |
| Ball closure = exact Mie, static                           | $2 \times 10^{-10}$ ; cube's bulk and shear average $10^{-8}$     |
| Ball closure vs Mie, dynamic, Eq. (25)                     | symbolic; numerical Mie $5 \times 10^{-4}$ ; $(ka)^3$ terms exact |
| Self term cancels, Eq. (27)                                | $4 \times 10^{-16}$                                               |
| Package $T$ = closure statically                           | $2 \times 10^{-5}$ at $k_S h = 0.01$                              |
| Second-order convergence                                   | order 2.000, three frequencies                                    |
| Closed-form error, Eq. (28)                                | $n = 1$ –16, to $10^{-4}$ of the error                            |
| First-moment voxel, Eq. (30)                               | order 4.00, three frequencies                                     |
| First-moment voxel, oblique, Eq. (33)                      | order 4, P and S, $20^\circ$ and $40^\circ$                       |

## 720 **Data availability**

721     The Mathematica notebooks and Python scripts that derive and check ev-  
722     ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)  
723     5281/zenodo.23003241 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)  
724     MultipleScatteringCalculations.

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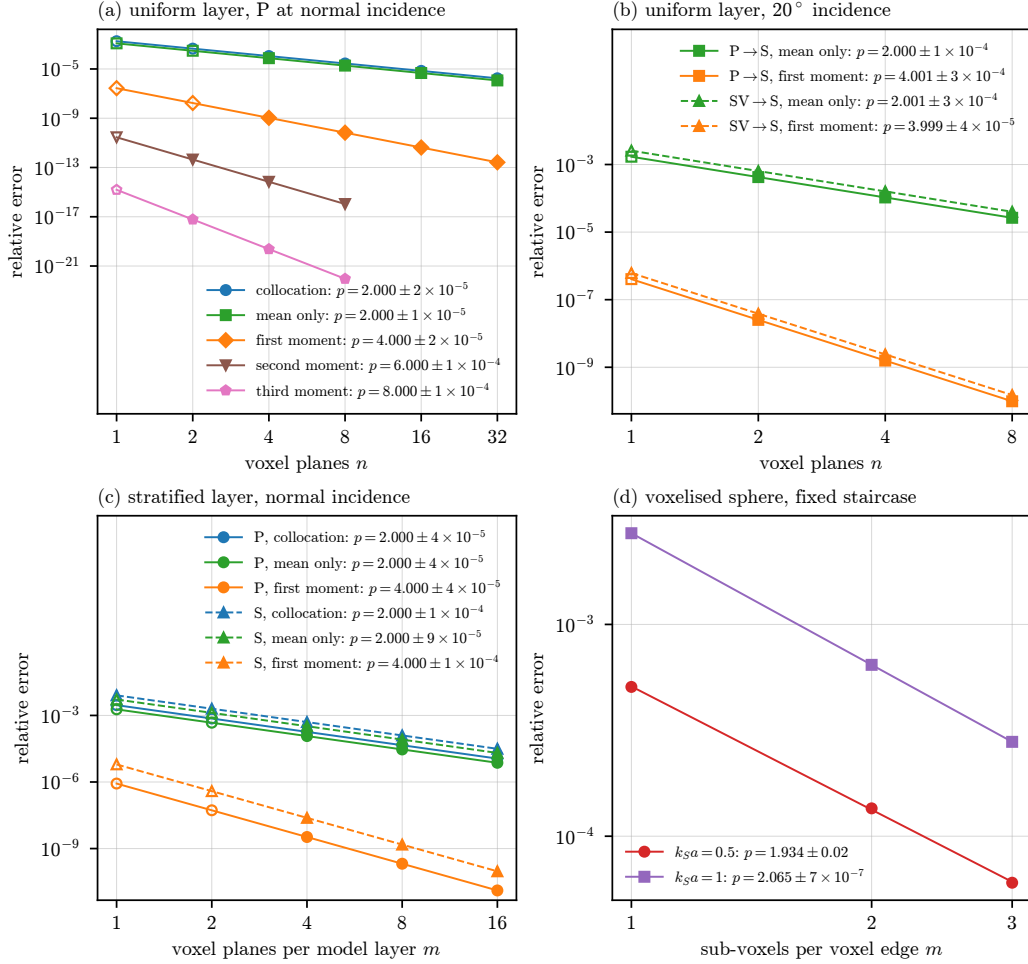


Figure 5: Convergence of every test, with least-squares regression lines. Each line fits  $\log e = c - p \log n$  to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order  $p$  and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree  $p = 1, 2$  and  $3$  (Eqs. (30) and (33)). (a) The uniform layer, P reflection at normal incidence ( $D = 2$  m,  $\omega = 300$  rad s $^{-1}$ , the contrast of §7), fitted from  $n = 4$ ; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from  $n = 2$ . (b) The uniform layer at 20°, fitted from  $n = 2$ : the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number  $m$  of voxel planes per model layer, fitted from  $m = 4$ . (d) The voxelised sphere of §10: the far-field difference between the  $n_{\text{sub}} = 8$  staircase with every voxel split into  $m^3$  sub-voxels and the staircase field extrapolated to  $m \rightarrow \infty$ , fitted over  $m = 1, 2, 3$ . That limit is extrapolated from the same three levels, so panel (d) checks consistency; its fitted orders differ slightly from those of Table 8 (1.99 and 2.06), which are measured from the changes between levels.

*Alt text:* Four log-log panels of relative error against refinement, each showing data